A precision platform under floor motion

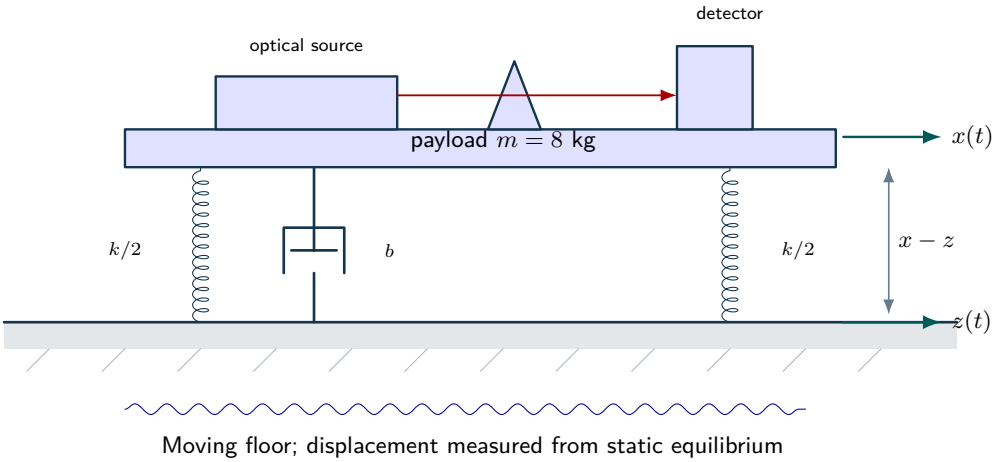
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

An optical platform has mass $m = 8\,\text{kg}$ and spring stiffness $k = 3200\,\text{N/m}$. Compare damping ratios $\zeta = 0.08$ and 0.40. Derive transmissibility from floor coordinate z to absolute platform coordinate x, find the isolation threshold and compare attenuation at $\omega = 2\omega_n$. Also find the displacement produced by a static 1 N force.

2 Physical system



An optical payload rests on parallel spring and damping elements. The two springs each have stiffness $k/2$, giving total k. Both the floor and payload move; spring extension and damping velocity are relative quantities. Weight is balanced at the static equilibrium about which the model is written.

3 Relative forces, absolute displacement

Let x be platform displacement and z floor displacement. The spring and damper act on relative motion:

$$m\ddot{x} = -b(\dot{x} - \dot{z}) - k(x - z) + f.$$

Thus the floor and force inputs have different numerators:

$$\frac{X}{Z} = \frac{bs + k}{ms^2 + bs + k}, \qquad \frac{X}{F} = \frac{1}{ms^2 + bs + k}.$$

With $m = 8$ kg and $k = 3200$ N/m, $\omega_n = 20$ rad/s and $b = 2\zeta\sqrt{mk} = 320\zeta$ N s/m. The two damping values are 25.6 and 128 N s/m.

4 Transmissibility and isolation

For $r = \omega/\omega_n$, the squared displacement transmissibility is

$$|X/Z|^2 = \frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}.$$

It is below one precisely when $(1 - r^2)^2 > 1$, or $r > \sqrt{2}$ for positive frequency. Damping does not move this crossing, but it changes resonance and high-frequency transmission.

5 A design tradeoff

At $r = 2$, use $\sqrt{[1 + 16\zeta^2]/[9 + 16\zeta^2]}$. Greater damping lowers resonance but increases transmission in the isolation band. A static 1 N force produces $x = 1/k = 0.0003125$ m irrespective of damping.

6 Limits

The floor is an imposed displacement, not a second independent mass. Mount stiffness and damping are ideal constants; travel limits, friction, rotations and material nonlinearities are excluded.

7 What to learn from the experiment

Reducing a resonance peak and improving high-frequency isolation are different design objectives. The transfer function must reflect which physical input is applied.

8 System sketch

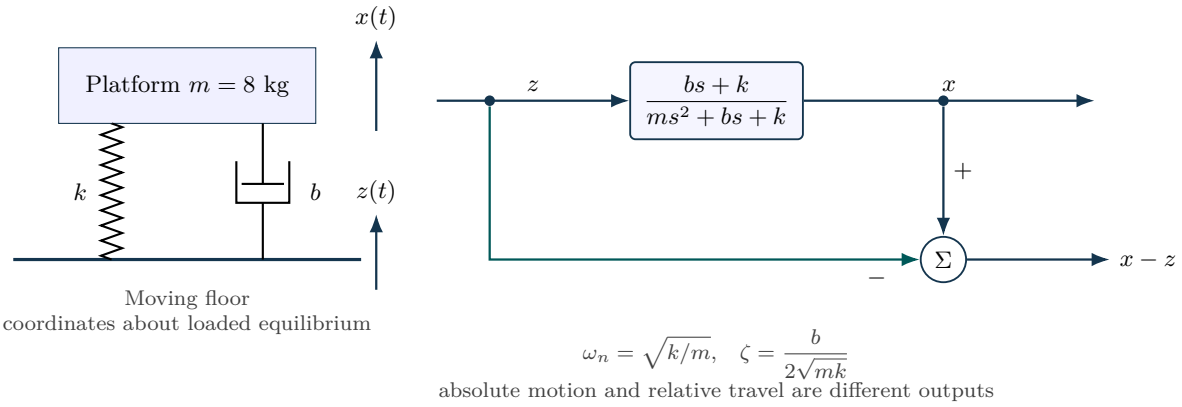
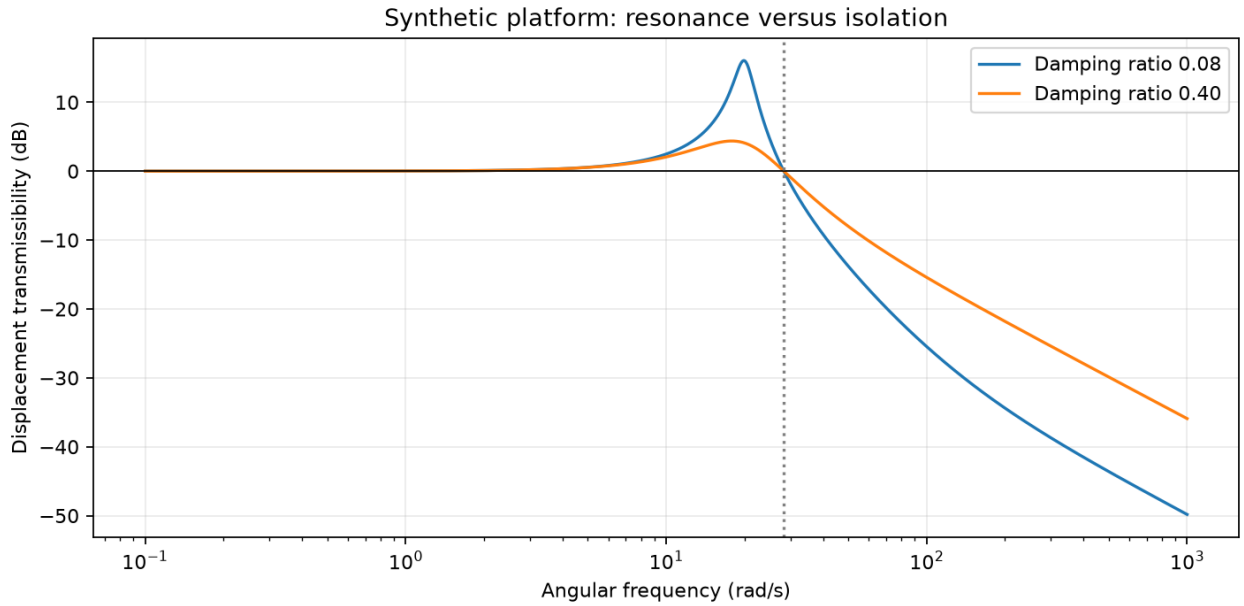


Figure 1. Base excitation drives absolute motion; subtracting the floor coordinate gives relative travel. The spring and damper act in parallel.

9 Executed computational evidence

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zeta=0.08, b=25.600 N s/m, transmissibility at 2 wn=0.348010
zeta=0.40, b=128.000 N s/m, transmissibility at 2 wn=0.554940
Isolation threshold: 28.284271 rad/s; static compliance: 0.00031250 m/N
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10 Method references

The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.